

With the help of functions ported from other open source packages, R handles signal-processing tasks pretty decently. These could be filter design or ECG signal analysis; researchers across the globe make efficient use of this tool for their applications.

As a signal-processing tool. "Bulk of R's basic signal-processing capability comes from the signal package that was ported over from the open source project Octave," points out Joseph Rickert in his blog titled 'R and Signal Processing.' The so-called signal package can be used to perform signal-processing functionalities including filtering, filter generation, resampling, interpolation and visualisation of filter models. These models are quite similar to the ones in MATLAB and hence anyone who has mastered the latter can easily switch over to this open source alternative.

When it comes to statistical analysis, time-series capabilities of R are superior to proprietary software like MATLAB or an open source rival like SAS. There are wrappers for MIT package for Fast Fourier Transforms called FFTW, dynamic linear modelling filter function based on singular value decompositions for Kalman filtering for Maximum Likelihood and Bayesian dynamic linear models.

R gives the freedom to users to explore Wavelets. They can make use of filters, transforms and multi resolution analysis from the wavelets package. For de-convolution on noisy signals, users can utilise WaveD transform. waveslim and wavethresh are other advanced wavelet signal-processing packages bundled in R.

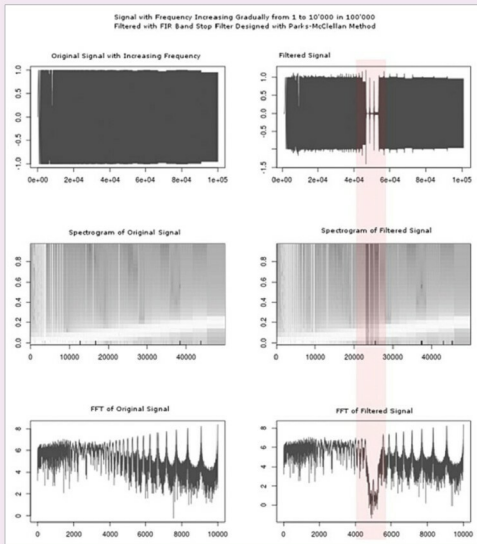


Fig. 3: Spectrogram and FFT of original and filtered signal (Courtesy: www.joyofdata.de)

For biomedical signal processing. Biomedical engineers have made effective use of the tool to adapt to various biomedical signal-processing tasks. Matt Shotwell has developed a reproducible R script for analysis of ECG signals using a windowed (Blackman) sinc low-pass filter. For eliminating high-frequency noise above 30Hz, a low-pass filter is applied to the signals at the first stage. In order to eliminate the slow wave that corresponds to respirations, the filter at a cut-off frequency of 1Hz is applied. You can find the reproduced ECG signals from raw data in the image alongside.

A language with unlimited possibilities

With more than 7000 additional packages extending the functionalities to 2000 applications, R has emerged as a statistical-computing environment with unlimited possibilities. Users can utilise the code developed by others in the open source community to adapt to their application. Hopefully, we can expect more communications and signal-processing applications to be actively developed on R in the near future. **EFY**

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